

AWS Operations Intelligence AI

Product Vision & Feature Requirements Document

Introduction

Now let's stop thinking like engineers and start thinking like product owners.

The biggest risk for this project is building:

"Yet another dashboard."

Nobody will care.

Especially when they already have Splunk.

What Are We Actually Building?

We are building:

AWS Operations Intelligence AI

Think of it as:

```
AWS Metrics + Splunk Data
              +
Enterprise Context
              +
AI Reasoning
              =
Operational Intelligence
```

The AI's job is not to show data.

The AI's job is to:

```
Observe
Analyze
```

Explain
Predict
Recommend
Alert

24/7.

Vision Statement

Imagine a new employee joining the company.

Instead of learning:

- Splunk
- CloudWatch
- AWS Console
- Reports
- Dashboards

They simply ask:

"What happened in AWS yesterday?"

And the AI answers.

That is the vision.

Phase 1 Features

Feature 1: AI Executive Summary

Every morning.

AI automatically generates:

AWS Consumption Summary

Yesterday:

- Glue executions increased 18%
- Lambda remained stable

- S3 storage grew 4%
- No critical anomalies detected

Audience

- Managers
- Directors
- Leads

This is huge value.

Feature 2: AI Trend Analysis

Instead of:

```
Glue Executions
Jan = 1000
Feb = 1200
Mar = 1800
```

AI explains:

```
Glue usage increased 80%.

Primary driver:
Analytics team ETL expansion.

Trend appears healthy and expected.
```

This is intelligence.

Feature 3: AI Anomaly Detection

AI continuously checks:

```
Glue
Lambda
SES
S3
Redshift
```

Detects:

```
Abnormal Growth
Abnormal Failures
Abnormal Usage
```

Example:

```
Lambda invocations increased 400%.

This exceeds historical growth patterns.
```

Feature 4: AI Recommendations

After detecting issues:

AI suggests:

```
Investigate Team A deployment.

Review Lambda retry settings.

Review Glue job failures.
```

Feature 5: Natural Language Query

User asks:

```
Which service grew the most this month?
```

AI answers.

User asks:

```
Show Redshift trend for last 90 days.
```

AI answers.

User asks:

Why did Glue executions increase?

AI explains.

Phase 2 Features

Now AI becomes more proactive.

Feature 6: Incident Intelligence

Splunk generates:

Critical Alert

AI investigates.

Instead of:

CPU > 90%

AI provides:

Root Cause Analysis

Cause:

New deployment at 11:05 AM

Impact:

Analytics pipelines delayed

Affected Services:

Glue, Redshift

This is where leadership starts seeing serious value.

Feature 7: AI Alert Prioritization

Many enterprises have:

500+
Alerts Daily

Nobody reads all of them.

AI categorizes:

Critical
High
Medium
Low

and explains why.

Feature 8: Stakeholder Reports

Different people get different reports.

Engineer

Technical details.

Manager

Operational summary.

Director

Business impact summary.

Same data.

Different AI-generated narratives.

Phase 3 Features

Now AI becomes an operations assistant.

Feature 9: Forecasting

AI predicts:

Next Week
Next Month
Next Quarter

Examples:

Lambda usage expected to increase 25%.

S3 growth likely to exceed storage threshold in 60 days.

Feature 10: Capacity Planning

AI warns:

Current growth trend suggests Redshift cluster scaling may be required within 45 days.

Feature 11: Cost Intelligence (Optional)

AI explains:

Glue costs increased due to ETL expansion.

S3 costs driven by archive retention.

Phase 4 Features

This is where the system becomes a true AI Operations Engineer.

Feature 12: AI Copilot

User asks:

What should I investigate today?

AI responds:

1. SES bounce rate increased.
2. Glue failures increased.
3. Redshift query latency worsening.

Feature 13: Daily AI Briefing

Every morning.

AI sends:

Good Morning.

Key Observations:

- 3 anomalies detected
- No critical incidents
- Storage growth accelerating
- Forecast indicates capacity review needed

Feature 14: Operations Knowledge Assistant

User asks:

Why is Lambda usage growing?

AI explains.

User asks:

What happened during last week's incident?

AI explains.

What Makes This Product Different?

Not:

Monitoring
Dashboards
Charts
Alerts

Splunk already does those.

Instead:

AI Explanation
AI Summaries
AI Recommendations
AI Forecasting
AI Incident Analysis
AI Operational Intelligence

Product Tagline

"An AI-powered AWS Operations Intelligence Platform that continuously analyzes enterprise cloud consumption, detects anomalies, explains trends, predicts future risks, and delivers actionable insights to engineering and leadership teams."

That is the product we are building.

The dashboard is just a screen.

The AI reasoning engine is the real product. 🚀
